

Using spatial ecology to examine above and belowground interactions on a reclaimed aspen stand in northern Alberta

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ABSTRACT

Examining the spatial interactions between above and belowground components of terrestrial ecosystems can give meaningful insight into the ecological processes happening at different scales. Understanding spatial dependence in these processes may help to evaluate reclamation success which is crucial for future management of such areas. The aim of this study was to measure the spatial patterns of soil biogeochemical properties in a young aspen stand reclaimed after oil sands extraction and to evaluate how the patterns were related to nutrient availability. Samples were collected from a 14-year old reclaimed site using a spatially explicit protocol with a minimum resolution of 0.5 m. Field-measured variables included forest floor depth and mass, tree cluster (canopy overlap), distance to nearest tree, and resin available nutrients. Soil microbial properties including microbial biomass C and N, basal respiration, and extracellular enzyme activity were measured during an eight-week laboratory incubation experiment. Geo-statistics were applied to examine the spatial patterns and model the space effect. A fine scale (<10 m) spatial pattern was found for the majority of stand variables, soil microbial properties, and available macronutrients (N, P, S and base cations). Macronutrients such as N, P and S availability had a fine scale cyclic spatial association with soil microbial properties, with an 8–10 m oscillation, which indicated belowground control on these nutrients. Spatial regression models also suggested a stronger microbial influence on the availability of these nutrients when compared to stand characteristics. However, stand characteristics exhibited significant control on base cations and micronutrient availability through the effect of forest floor depth and tree clustering. Although nutrient availability showed strong spatial relationships with belowground processes in the studied reclaimed site, similar relationships with aboveground properties appeared to be weak, and might require further time to develop.

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1. Introduction

Heterogeneity in ecosystem processes may arise from interactions between above and belowground components (Kardol and Wardle, 2010). There are both positive and negative feedback loops through which one component influences the other, and sometimes these forces work together under the same spatio-temporal scale to generate specific ecosystem patterns (Wardle et al., 2004). Above and belowground linkages at one level of ecosystem organization can transcend to a higher level of organization; for example the interaction between soil microbes and plant functional traits can directly shape community level biodiversity, which has the potential to affect ecosystem level C and nutrient fluxes (Kardol and Wardle, 2010; Van Der Heijden et al., 2008). Aboveground properties such as tree distribution, tree cluster, and forest floor depth can directly or indirectly affect variability in soil microbes, including composition and function (Wardle, 2002; Weber and Bardgett, 2011). Trees can influence the spatial distribution of under-story species and belowground biota by modifying resources

such as light, moisture, and by their variable rate of litter input and nutrient uptake from soil (Saetre, 1999; Wardle, 2002; Weber and Bardgett, 2011; Turner et al., 2011). Belowground organisms can also significantly influence aboveground vegetation dynamics and distribution. Changes in the soil microbial community can favor plant associations of a certain type by directly modifying organic matter decomposition pathways, and altering belowground nutrient dynamics (Bradford et al., 2002; John et al., 2007).

Microbes are very important biogeochemical agents in marginal or heavily disturbed soil, where they are responsible for initiating ecosystem development and creating resource heterogeneity (DeGroot et al., 2005; Smithwick et al., 2012; Van Der Heijden et al., 2008). Stolp (1988) proposed that microbially driven biogeochemical heterogeneity in soils is determined by the availability of microbial resources and abiotic reaction conditions. Soil microbial control on nutrient availability, therefore, could be stronger at the initial stage of stand development in disturbed ecosystems (Wardle, 2002), when the disturbance creates large amounts of labile soil organic matter, as is the case after wildfire (Certini, 2005; Choromanska and DeLuca, 2002; Fernández et al., 1997) and after tillage in agriculture (Marriott and Wander, 2006). Strong microbial influence on vegetation dynamics and nutrient

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biogeochemistry at the stand initiation phase has also been reported in human disturbed ecosystems such as in ecosystems recovering from surface mining and harvesting (Holmes and Zak, 1999; Ingram et al., 2005; Mummey et al., 2002). In nutrient limited northern ecosystems such as the boreal, soil microbial influence on the spatial distribution of nutrients is likely to have similar magnitude as plant contributions since the aboveground and belowground feedback loops in these ecosystems are tightly linked through the accumulation of organic matter (Eskelinen et al., 2009; Van Der Heijden et al., 2008). Disturbance, especially surface mining and reclamation with novel soil substrates, can reshuffle the established spatial structure in the aboveground and belowground processes, and these patterns might take decades to recover as seen in abandoned agricultural fields (Flinn and Marks, 2007).

In this study, we seek to tie patterns of nutrient availability to above and belowground properties with spatially explicit sampling and geostatistics. Fig. 1 and the following outline our specific hypotheses with the objective of: 1) examining spatial predictability of above and belowground properties, 2) isolating the microbial influence on nutrient availability, and 3) examining tree influence on microbial activity.

1.1. Hypothesis 1

Young oil sands reclaimed sites usually have less heterogeneity in terms of geomorphological features, including: micro-topography, slope, and sub-surface materials (Fig. 1). Low heterogeneity in juvenile reclaimed ecosystems has been reported in previous studies (e.g. Nyamadzawo et al., 2008; Shukla et al., 2005). Heterogeneity might develop in such reclaimed ecosystems over time through continuous organic matter input (Boerner et al., 1996, 1998), modification of soil physico-chemical conditions (Shukla et al., 2007), and proliferation of a diverse soil microbial community (Anderson et al., 2008). However, the amount of time it might take for heterogeneity to develop has not been well documented. The most significant aboveground changes in forest ecosystems happen during the canopy closure phase of stand development. Natural fire disturbed aspen ecosystems usually reach canopy closure by 7 to 11 years after initiation (Petersen and Petersen, 1992; Valverde and Silvertown, 1997). During this time forest floor litter and stand structure will develop predictable patterns that may increase with tree gap or cluster formation (Cumming et al., 2000). Development

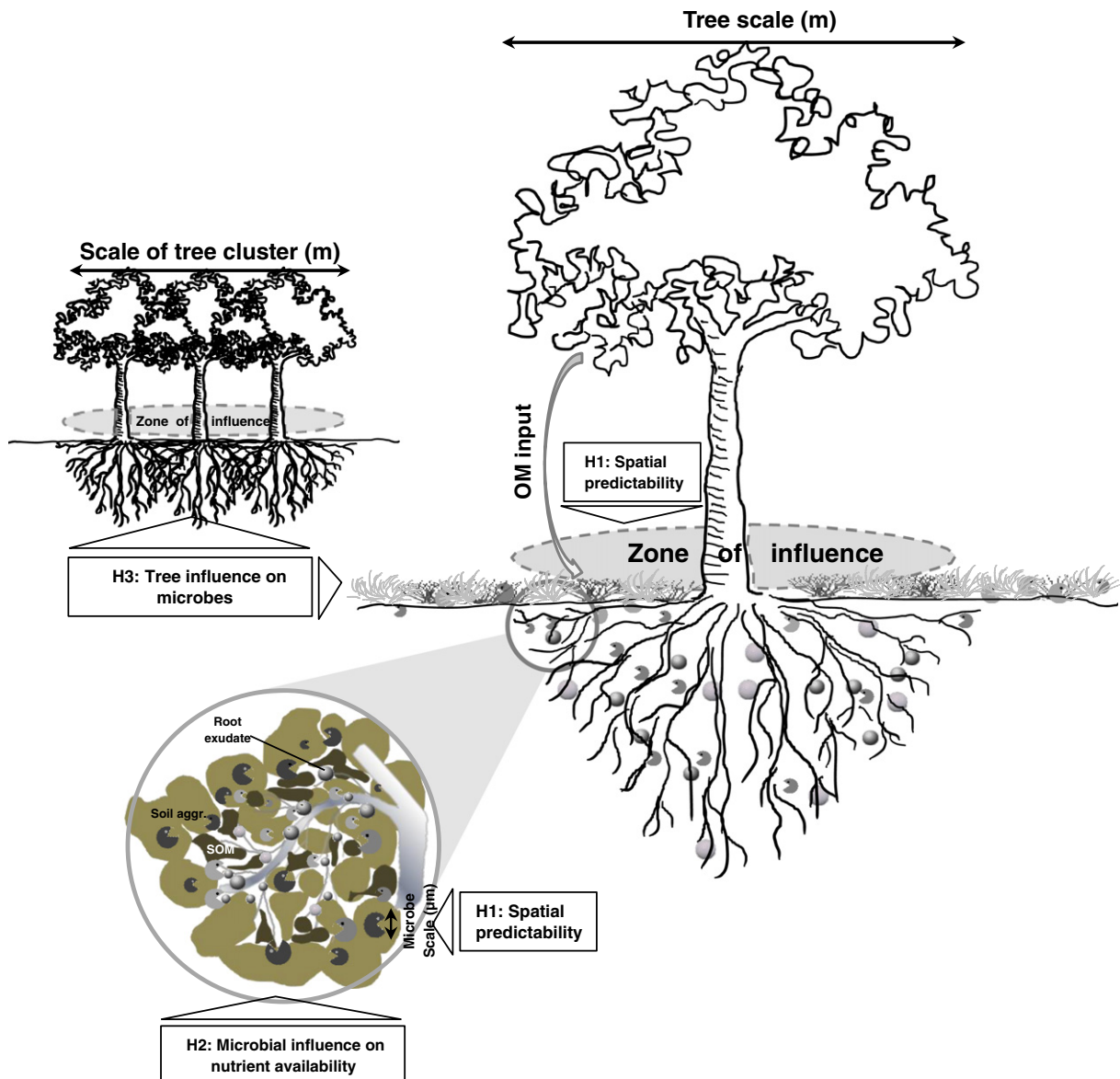


Fig. 1. Schematic diagram showing different parts of the working hypotheses.

of heterogeneity in soil nutrients and microbial properties has also been observed within ten years of natural disturbance, such as wildfire in boreal ecosystems (Lavoie and Mack, 2012; Smithwick et al., 2005). Therefore, we assume that spatial heterogeneity in above and belowground properties might develop in oil sands reclaimed ecosystem, and examined this in a 14 year old reclaimed site using a semi-variogram approach.

1.2. Hypothesis 2

Spatial associations with available nutrients occur mostly at the scales of enzymatic reactions necessary to break litter polymers into smaller molecules (Allison, 2005), microbial metabolism (Stoyan et al., 2000), nutrient uptake (Selles et al., 1999), and their combined effect (Schimel and Chapin, 2006) (Fig. 1). The quantity and nature of extracellular enzymes produced by microbes depend mostly on the biochemical characteristics of the native organic matter. Thus, the relative activity of enzymes associated with decomposition should better represent mineralization and nutrient release than other less specific indices, such as total microbial count (Decker et al., 1999). Soil enzymes and microbial metabolism have been shown to be related to nutrient mineralization at variable hierarchical scales in ecosystems recovering from natural disturbances. The scale of such spatial association ranges from <1 m to 10 m (Goffman et al., 2009; Smithwick et al., 2005). We believe that nutrient availability in oil sands reclaimed ecosystem would have a fine scale spatial association with microbial properties. We hope to tease apart these belowground interactions using cross-variogram analysis and spatial regression.

1.3. Hypothesis 3

Trees can directly or indirectly influence above- and belowground processes through modifications of site condition (Fig. 1). The major influence of trees on soil properties results from modifications in litter quality, soil temperature, and hydrological regimes, and all these in turn shape the belowground microbial assemblage (Schimel and Chapin, 2006). Consequently, the modifications in decomposer community could be directly linked to nutrient availability (Hypothesis 2). Several studies (e.g. Zinke, 1962; Kuuluvainen et al., 1993) have been conducted which provide evidence for a “zone of influence” under single tree or tree cluster (Fig. 1). The spatial range at which individual trees or tree clusters influence belowground processes usually occurs at the scale of either the canopy or root distribution (Saetre, 1999; Saetre and Bååth, 2000; Ettema and Wardle, 2002). Competition is also responsible for creating patterns in planted stands; such spatial range has been shown to be 1–4.5 m (Fajardo and McIntire, 2007). Therefore, we expected to see a strong spatial association between tree cluster and soil microbial properties at ≥ 3 m (the maximum spacing between plants), and used cross-variogram analysis and spatial regression approach to tease these apart.

Re-establishing aboveground and belowground linkages should be the primary goal of reclaiming ecosystems that are disturbed by human resource extraction (Kardol and Wardle, 2010). Surface mining in the Athabasca Oil Sands Region (AOSR) of northern Alberta has created a unique opportunity to study the development of heterogeneity in these components. The legal mandate requires extraction companies to reclaim the disturbed landscape to the equivalent land capability that existed prior to disturbance and to selectively salvage soils for use in reclamation (Alberta Government, 2014). Recently, a land capability classification system (LCCS) (Leskiw, 1998) was used for measuring the potential of reclaimed sites to grow upland boreal forests. However, capability might not fully represent structural and functional compositions at different scales, and therefore mislead the ecological fidelity of reclamation (see Higgs, 1997; Kelly and Harwell, 1990). What we are proposing is that ecosystem heterogeneity (or spatial pattern analysis) can be used as a tool for measuring reclamation success/function, either

in combination with the LCCS or separately. To our knowledge, only one other spatial study has been done in the AOSR region, which did not account for the belowground spatial control on nutrient availability (see Sorenson et al., 2011a). Identifying fine scale heterogeneity in nutrient availability and their aboveground–belowground control mechanisms in novel, reconstructed ecosystem would enhance our understanding of ecosystem progression after severe disturbance, and at the same time, can assist in measuring reclamation success when compared with natural analogues.

2. Methods

2.1. Study area

The study site, a 14 year old aspen (*Populus tremuloides* Michx.) stand reclaimed in 1994, is located 40 km north of Fort McMurray, Alberta (57°05′00.9420″N 111°36′44.5654″W). The climate in this region is characterized by short summers and long cold winters. The mean monthly air temperature in the region ranges from -17.4 °C in January to 17.1 °C in July, with a mean annual temperature of 0.9 °C. The mean annual precipitation is 418 mm, with 283 mm occurring as rainfall during the growing season (Environment Canada, 2014). Details about reconstruction of soil profiles following oil sand mining can be found in Rowland et al. (2009) and Sorenson et al. (2011b). Briefly, soil profiles were reconstructed with surface salvaged organic and geologic materials removed during oil sands mining, and placed as a 10 cm cap of peat and 40 cm of mineral soil as a clean top layer (peat mineral mix; PMM). After placement of capping substrates, the site was offset disked and aspen seedlings were planted at a density of 1700 stem ha^{-1} with an average spacing of 2–3 m between trees. Complete fertilizer (N–P–K–S) was applied at a rate of 500 kg ha^{-1} with a blending ratio of 10:30:15:4 to favor the initial establishment of aspen seedlings.

2.2. Sampling design

Sample locations were established using a spatially explicit sampling protocol (Appendix Fig. 1). A random walk design (Underwood, 1997) was followed, where sampling locations and directions were fixed from a list of randomly generated numbers. A minimum of 30 paired points at each lag distance were selected in order to increase detectable spatial relationships (Journel and Huijbregts, 1978). With the random walk method, 15 initial points were located with an interval of 2–10 m between points, and at each of these, four points were established in cardinal directions at random distances ranging from 0.5 m to 2 m, giving a total of 75 sample points. The minimum spatial resolution of the current sampling protocol was 0.5 m.

All field measurements and sampling were done in August 2008. Key aboveground and belowground variables linked to nutrient availability were measured. Soil variables were selected from the areas that directly influence nutrient bio-geochemistry such as substrate quality, microbial activity and chemical environment (Ingram et al., 2005; Mummey et al., 2002). Plant root simulator (PRS™) probes (Western Ag Innovations Inc., Saskatoon, SK, Canada) were used to measure nutrient availability under field conditions. Cation and anion probes were installed vertically at each sampling point with the top of the probe membrane at the interface of the forest floor and PMM, and left for 12 weeks. Upon retrieval, probes were extracted with 0.5 M HCl and elutes were analyzed for nutrients. Inorganic nitrogen (NH_4^+ and NO_3^- ; TIN), and phosphate were analyzed colorimetrically using a segmented flow Autoanalyzer III (Brand and Lubbe, Inc., Buffalo, NY). Potassium, sulfate, calcium, magnesium and other micronutrients were quantified by inductively-coupled plasma (ICP) spectrophotometry (PerkinElmer Optima 3000-DV, PerkinElmer Inc., Shelton, CT).

Forest floor and PMM samples were collected using bulk density cores (15.2 cm and 5 cm diameter, respectively). PMM samples were collected to a depth of 5 cm. Forest floor depth (FFdepth) was recorded

at each point and forest floor mass (FFmass) was calculated on an areal basis from bulk density (Mg ha^{-1}) determined by drying the samples at room temperature (23°C). Finally, location of each tree within 10 m of a sampling point was surveyed using a Vertex III and Transponder T3 with an automatic level. Distance to nearest tree and number of overlapping canopies (stated here as tree cluster) on each sampling point was computed.

2.3. Incubation experiment

PMM samples were brought back to the laboratory, sieved (2 mm) and air dried. Total organic C (C) and total N (N) were measured by the Dumas Combustion Method using a Costech 4010 Elemental Analyzer System (Costech Analytical Technologies Inc., Valencia, CA, USA). Soil pH was measured with a 2:1 ratio of 0.01 M CaCl_2 (Kalra and Maynard, 1991).

A 45 day lab incubation experiment was conducted to characterize the key microbial and biochemical properties of these PPM samples. Before incubation, samples were mixed with 20% quartz sand (pH 7.0) to ensure aeration and uniform moisture distribution. Sample moisture was adjusted to 60% of field capacity determined by a pressure plate experiment (at -33 kPa pressure). Approximately 100 g of soil–sand mix was pre-incubated for 3 weeks at a constant temperature of 25°C to avoid the sudden flush of soil respiration and to re-establish microbial function. During this period, CO_2 generation was monitored using gas chromatography (HP 5890, Supelco 8 ft $\times$ 1/8" O.D. stainless steel column, matrix 80/100 HayeSep Q) at 2–3 day intervals until a stable respiration point was reached. After the pre-incubation, soils were mixed with ion exchange resin (J. T. Baker, mixed-bed exchange resins) to adsorb mineralized nutrients. The experimental incubation was then conducted for 4 weeks. Sub-samples (15 to 20 g) were collected at the beginning and end of incubation, and extracted with 2 M KCl (1:3) for inorganic N (TiN_{inc} , $\text{NH}_4^+_{\text{inc}}$ and $\text{NO}_3^-_{\text{inc}}$). Gravimetric moisture content was adjusted by taking jar weight at every 3 to 4 days during the incubation experiment. Basal respiration was measured for three successive days before the end of the incubation using gas chromatography. Finally, samples were analyzed for different microbial properties at the end of the incubation. Measured properties included microbial biomass C and N (MBC and MBN), dissolved organic C and N (DOC and DON), hexose sugar complex (HSC), and extracellular enzyme activities related to C, N, P and S mineralization.

Microbial biomass C and N were measured according to the chloroform fumigation extraction method (Vance et al., 1987) using 20 g soil, extracted in 0.5 M K_2SO_4 with a ratio of 1:2. Dissolved organic C and N were measured on the unfumigated samples using a Shimadzu TOC-V/TN analyzer (Shimadzu Corp., Kyoto, Japan). Soluble hexose sugar complex (HSC) was measured in unfumigated soil extraction by Anthrone reactive C method as described in DeLuca and Keeney (1993). The amount of pentose and other sugar complexes (PSC) was calculated by taking the difference between DOC and HSC. Microbial quotient (respiration per unit biomass; qCO_2) was calculated by dividing the BR with MBC. Ammonium and nitrate ($\text{NH}_4^+_{\text{inc}}$ and $\text{NO}_3^-_{\text{inc}}$, respectively) from the incubation experiment were measured colorimetrically on a SmartChem Discrete Wet Chemistry Analyzer (Westco Scientific Instruments, USA).

Extracellular enzyme activities were measured using microplate assays as described by Sinsabaugh et al. (2003). The activities of β -1, 4-glucosidase (β -glucosidase, E.C. 3.2.1.21), N-acetyl- β -glucosaminidase (NAGase, EC 3.2.1.30), acid phosphatase (E.C. 3.1.3.2) and sulfatase (E.C. 3.1.6.1) were measured using 4-methylumbelliferyl (MUB) derived substrates. A 200 μl volume from a soil suspension (1 g soil homogenized in 100 ml 0.1 M, pH 5 sodium acetate), and 50 μl of 200 μM substrate were pipetted into each sample well of a 96 well plate. Fifty microliters of 4-methylumbelliferone (10 μM) plus 200 μl of acetate buffer were used in the reference standard wells. Quench control wells received 200 μl of soil suspension plus 50 μl of MUB standard.

Each sample was replicated eight times per enzyme assay. Microplates were incubated at 20°C in the dark for up to 3 h depending on the assay (a 2 hour incubation was used for acid phosphatase). Fluorescence was measured at 365 nm excitation and 460 nm emission, using a microplate spectrophotometer (Synergy HT, BioTek, Winooski, VT). Activity rates (μmol of converted substrate g^{-1} soil h^{-1}) were calculated on an oven dry basis.

2.4. Spatial statistics

We found significant spatial patterns in the data, therefore we could not consider our samples to be spatially independent, making parametric statistics of limited use because of inflated degrees of freedom (Fortin and Dale, 2005; MacKenzie et al., 2008). Failing to account for any significant spatial structure in data has been demonstrated to have resulted in dramatically different conclusions on ecosystem processes (Fortin and Dale, 2005; Keitt et al., 2002). Therefore, spatial statistics were used to account for the residual spatial pattern in the data.

2.4.1. Spatial autocorrelation

The coefficient of variation (CV) was used as a measure of global variation in the soil and stand properties, and a semi-variogram approach was used to detect spatial autocorrelation (local variation). Individual variables were first tested to check for any large scale directional trend, and, if found, trends were removed by polynomial functions using X and Y coordinates (Lloyd, 2010). Semi-variograms were calculated with a minimum lag spacing of 0.5 m and a maximum lag distance of 30 m. The nugget to total variance ratio was used to determine spatial dependency. A ratio of less than 25 was treated as weak, between 25 and 75 as moderate, and greater than 75 was considered strong spatial dependency (Cambardella and Elliott, 1994). Here, spatial dependency was expressed on a scale of 0 to 1, where 1 means perfect spatial dependence. Spatial autocorrelation at >10 m range (from semi-variogram) was termed as 'large scale', and all the spatial structure <10 m was termed as 'fine scale'. Trend surface analysis was done in openGeoDa package (Anselin, 2004). Semivariograms and ordinary Kriging maps were produced using ArcGIS Geostatistical analyst module (ArcGIS v 9.3) and GS+ version 9.0 (Robertson, 2008). The nearest-neighbor distribution ($G(r)$) was calculated for aspen trees using spatstat library (Baddeley and Turner, 2005) in R (R Development Core Team, 2011). The upper and lower envelopes of $G(r)$ were computed from 99 Monte Carlo simulations. Nearest-neighbor values above the upper envelope indicate tree aggregation, within the envelopes indicate random distribution, and below the lower envelope indicate regular distribution (Baddeley and Turner, 2005).

Cross-variograms were calculated to examine the spatial dependence between two variables, and to detect at which distance this association occurs. Assumptions of stationarity and isotropy also hold true for cross-variogram (Journel and Huijbregts, 1978; Yates and Warrick, 2002). Log and square root transformed variables were used in calculating cross-variogram to avoid any outlier effect. Positive cross-variance means a positive spatial association, whereas, negative cross-variance indicates spatial dissociation (McBratney and Webster, 1983). Equations for both semi-variogram and cross-variogram are given in the Appendix.

2.4.2. Spatial regression modeling

Spatial regression models (Anselin, 1988; Besag, 1974) were used to determine what factors were responsible for the variation in available nutrients, as well as to quantify their proportional contribution to this variation after accounting for spatial residuals. A two-step process combining both standard regression and geostatistical techniques was used. As spatial pattern of nutrient availability very much depends on the pattern of multitude of factors such as substrate quality (C:N and labile C), microbial activity (enzyme production and mineralization), and chemical environment (pH, moisture and temperature), we built the initial

spatial model with the variables that mostly represents the three categories. Soil properties and stand characteristics were used as covariates, and resin available nutrients were the dependent variables in the regression models. Spatial structures in the selected models were tested by ordinary least square regression (OLSR) with spatial error term and the spatial autoregressive model (SAR). Space term in the model can be regarded as a proxy of unmeasured variables (McIntire and Fajardo, 2009). SAR model was used for its ability to propagate local neighborhood effect in the final model, and to detect spatial autocorrelation due to any unmeasured variables (Keitt et al., 2002). These specific traits of the spatial model are important as the right combination of variables controlling nutrient availability in the novel ecosystem was unknown. SAR was only used when there was significant spatial autocorrelation detected in the model residuals, tested by calculating the global Moran's I (Anselin, 2004). Spatial regression analysis was done using Geoda (Anselin, 2004) and spdep packages (Bivand et al., 2011) in R program (R Development Core Team, 2011).

3. Results

3.1. Spatial autocorrelation

Spatial autocorrelation was found for all the measured soil properties and stand characteristics (Table 1). Most of the available macronutrients had fine scale (<10 m) spatial range (Table 2). No spatial autocorrelation was found for micronutrients except for Al and Zn. Aluminium had a strong spatial dependency whereas Zn displayed a much weaker relationship. Fine scale spatial range and high global variability (CV from 34 to 63%) were found in soil microbial properties. On the other hand, soil chemical properties such as C, N, and pH showed a spatial range exceeding 10 m with low global variability (CV from 2 to 21%). The spatial range of stand characteristics varied between 2.6 m (forest floor mass) and 4.2 m (tree cluster). Tree distribution showed an aggregated pattern from 0.75 m to 2 m (Fig. 2). The degree of spatial dependence ranged from 3 to 68% for nutrient availability, 55 to 90% for soil microbial properties, and 30 to 60% for stand characteristics. Most of the soil microbial properties had moderate spatial dependency (Table 1).

3.2. Spatial association

Spatial associations were detected between nutrient availability, microbial properties and stand characteristics as indicated by cross-

variance analysis (Fig. 3). Fine scale (<10 m) spatial association was found between N–P–S availability and microbial properties (Fig. 3a, b and c). NO_3^- inc. showed a positive association with MBC (7 m) as evident from the fitted variogram model (Fig. 3a line), but cross-variance also followed a cyclic pattern with maximum amplitude at 8 m, gradually decreasing to 12 m (Fig. 3a points). Such cyclic pattern was observed in most of the cross-variograms between N–P–S availability and microbial properties. For example, cross-variance between P availability, acid phosphatase and β -glucosidase had a cyclic pattern at every 10 m and the fitted variogram models indicated a spatial range <4 m (Fig. 3b). The similar cyclic pattern was also found in the cross-variance between S availability and metabolic quotient, although no spatial range was detected for the current scale of measurement (Fig. 3c). Potassium, Ca and Mg did not show any clear spatial association with microbial properties.

Spatial associations between microbial properties and stand characteristics showed either large scale (>15 m) or very weak pattern. For example, MBC and DOC showed negative association with tree clustering at 16 m and 21.1 m, respectively (Fig. 3d). Enzyme activities such as β -glucosidase and acid phosphatase, on the other hand, had very weak to random association with tree cluster. Cross-variance analysis between nutrients and stand characteristics also revealed spatial associations at larger scales (≥ 23 m) and did not show any cyclic pattern (data not shown). Bivariate correlation analysis (global correlation) between microbial properties and stand characteristics further confirmed the poor dependency between these variables (Appendix Table 1).

Spatial distribution of available nutrients was more similar to the spatial distribution of soil microbial properties as indicated by Kriging (Fig. 4). Hot and cold spots in P availability matched well with that of BR, phosphatase, and β -glucosidase activity. The TIN (both resin available and incubation) map had patch similarity with DON maps. The TIN_{inc} map also exhibited patch similarity with MBC, BR, and β -glucosidase activity. Sulfur availability did not show any distinct patch similarity with a single property, however, it had a mixed pattern similar to β -glucosidase activity and BR.

3.3. Spatial regression models

Both the separate and combined effects of above and belowground attributes were tested on nutrient availability. Models with predictor variables including stand characteristics and soil chemical properties

Table 1
The mean, coefficient of variation (%), spatial range, spatial dependence class and semi-variogram models fitted to soil properties and stand characteristics of a 14 year old aspen reclaimed site.

	Factors	Mean ($\pm$ SE)	CV (%)	Range (m)	Spatial dependence	Dependence class	Model	RMSE ^a	
Stand characteristics	FFdepth (cm)	2.1 $\pm$ 0.05	21.1	2.2	0.44	Moderate	Gaussian	1.10	
	FFmass (Mg ha ⁻¹)	40 $\pm$ 10	31.4	2.6	0.32	Moderate	Spherical	1.13	
	Tree cluster (canopy overlap)	1.3 $\pm$ 0.13	84.5	4.2	0.68	Moderate	Exponential	1.06	
	Treedist. (m)	0.85 $\pm$ 0.06	57.5	3.2	0.62	Moderate	Gaussian	1.04	
Soil properties	Microbial	MBC ($\mu\text{g g}^{-1}$ soil)	178.3 $\pm$ 9.41	45.7	7.7	0.64	Moderate	Gaussian	1.07
		MBN ($\mu\text{g g}^{-1}$ soil)	17.1 $\pm$ 1.57	62.5	8.5	0.71	Moderate	Gaussian	1.08
		HSC ($\mu\text{g g}^{-1}$ soil)	41.4 $\pm$ 2.44	51.2	8	0.90	Strong	Exponential	1.09
		PSC ($\mu\text{g g}^{-1}$ soil)	154.8 $\pm$ 7.69	43.0	18.6	0.76	Strong	Circular	0.99
		DON ($\mu\text{g g}^{-1}$ soil)	48.3 $\pm$ 1.97	35.5	4.7	0.67	Moderate	Gaussian	1.13
		BR (mg CO ₂ -C kg ⁻¹ soil day ⁻¹)	4.1 $\pm$ 0.26	54.6	6.7	0.73	Moderate	Circular	1.10
		qCO ₂	0.02 $\pm$ 0.001	40.1	12.3	0.60	Moderate	Circular	1.05
		β glucosidase ($\mu\text{mol g}^{-1}$ soil h ⁻¹)	0.12 $\pm$ 0.005	35.5	5.8	0.55	Moderate	Circular	1.12
	Chemical	NAGase ($\mu\text{mol g}^{-1}$ soil h ⁻¹)	0.07 $\pm$ 0.003	34.5	4.0	0.79	Strong	Exponential	1.25
		Phosphatase ($\mu\text{mol g}^{-1}$ soil h ⁻¹)	0.08 $\pm$ 0.004	43.5	5.3	0.52	Moderate	Gaussian	1.09
		Sulfatase (nmol g ⁻¹ soil h ⁻¹)	1.47 $\pm$ 0.06	35.2	7.2	0.61	Moderate	Circular	1.13
		C (g kg ⁻¹ soil)	92.4 $\pm$ 2.26	21.2	23	0.50	Moderate	Gaussian	1.06
		N (g kg ⁻¹ soil)	4.1 $\pm$ 0.12	24.8	19	0.47	Moderate	Gaussian	1.04
		C:N	22.5 $\pm$ 0.22	8.4	13.5	0.40	Moderate	Gaussian	0.98
		pH	6.7 $\pm$ 0.02	2.2	13	0.16	Weak	Gaussian	0.97

^a RMSE = root mean square error.

Table 2

The mean, coefficient of variation (%), spatial range, spatial dependence class and semi-variogram models fitted to the available nutrients in a 14 year old aspen reclaimed site measured using PRS probes ($\mu\text{g } 10 \text{ cm}^{-2}$) and lab incubation experiment.

	Factors	Mean (SE)	CV (%)	Range (m)	Spatial dependence	Dependence class	Model	RMSE
Macro-nutrients	$\text{NH}_4^+ \text{ inc.}^a$	0.35 ± 0.03	70.9	6.8	0.66	Moderate	Gaussian	1.06
	$\text{NO}_3^- \text{ inc.}$	30.1 ± 2.42	69.7	6.5	0.68	Moderate	Gaussian	1.05
	NH_4^+	2.55 ± 0.31	104.3	5.5	0.13	Weak	Circular	0.98
	NO_3^-	7.83 ± 1.47	162.8	2.0	0.28	Moderate	Circular	0.81
	P	4.78 ± 0.35	63.4	3	0.28	Moderate	Gaussian	0.98
	K	103.2 ± 7.12	59.8	4.2	0.07	Weak	Exponential	0.97
	S	631.8 ± 22.4	30.7	5.7	0.21	Weak	Gaussian	0.97
	Ca	2989.8 ± 36.5	10.5	25	0.03	Weak	Circular	0.95
	Mg	390.7 ± 6.23	13.8	ND	ND	ND	ND	ND
	Fe	18.67 ± 1.36	63.1	ND	ND	ND	ND	ND
Micro-nutrients	Zn	2.94 ± 0.19	57.9	14	0.15	Weak	Exponential	1.05
	B	3.08 ± 0.20	56.8	ND	ND	ND	ND	ND
	Mn	0.63 ± 0.05	72.7	ND	ND	ND	ND	ND
	Al	79.4 ± 2.85	31.1	10.3	0.79	Strong	Gaussian	0.99
	Cu	0.41 ± 0.01	20.0	ND	ND	ND	ND	ND

ND = Not detected.

^a inc. = from incubation experiment.

(pH, C and N) indicated a significant space effect for TIN, S, P, Al and Cu availability. However, after incorporating soil microbial properties in the model, the significant space effect disappeared, indicating a sufficiency of predictor variables in the model at the scale of measurement (Tables 3 & 4). Nitrogen availability was mostly controlled by MBC (positively) and pH (negatively), whereas β -glucosidase activity and soil C were the most significant controlling factors for P availability. HSC and DON were found to be the most significant factors for S availability. No significant spatial structure was detected for Ca, K, Mn and B when only stand characteristics were in the regression model; however, including soil microbial properties in the model lowered the AIC and almost doubled the R^2 . Stand properties, specifically forest floor depth and soil pH, were significant predictors of Mg, Ca, K, Fe, Zn and Al. Significant microbial control through BR, β -glucosidase, and phosphatase activity was found for micronutrients such as Mn, B and Cu. Aluminium

and $\text{TIN}_{\text{inc.}}$ both had significant spatial structure in the regression model even after incorporating stand characteristics and microbial properties (Appendix Table 2).

4. Discussion

4.1. Spatial pattern of soil and stand properties

Fine scale spatial patterns in nutrient availability, soil microbial properties, and stand characteristics indicate that aboveground and belowground processes in this reclaimed aspen ecosystem have developed some heterogeneity. This finding supports our first hypothesis that aspen stands should develop spatially predictable variability in soil and stand attributes; however, the relationships are not as strong as we would expect after 14 years of recovery from oil sands reclamation. Heavily disturbed and managed systems (e.g. agricultural sites) tend to have either large scale or no spatial pattern in soil properties (Fraterrigo et al., 2005; Robertson et al., 1993). We believe that at initiation oil sand reclamation sites are comparable to agricultural sites, because homogenization had happened at different stages during soil salvage and placement when the site was built. There might be a possibility that larger blocks of soil and organic materials might have retained some of the spatial variability from the donor sites, however, offset disking done after soil placement was very likely to have destroyed the ecological legacies that had created the variability. Over time, heterogeneity must develop in the reclaimed system through contributions of mass and energy from plants and soil microbes (Anderson et al., 2008; Shukla et al., 2007). Cambardella et al. (1994) reported >100 m spatial range of soil microbial biomass and N availability in a no-till agricultural field which had been previously managed through tillage. Flinn and Marks (2007) found that spatial homogenization of soil properties and altered vegetation composition remained as a legacy of former agricultural practices even after 100 years of reforestation. Lesschen et al. (2008), in semi-arid ecosystem, showed that recovery of spatial heterogeneity in vegetation and soil properties takes at least 40 years after abandonment of agriculture field. On the other hand, Banning et al. (2008) and Graham and Haynes (2004) reported the development of natural levels of variation in soil microbial properties 18 to 22 years after reclamation and attributed this to aboveground factors such as litter quality. Mummey et al. (2002) found a fine scale structure (<14 cm) in soil microbial community composition and N availability in a 22 year old reclaimed grassland and proposed that belowground controls, including plant root activity and nutrient foraging, had contributed to this heterogeneity.

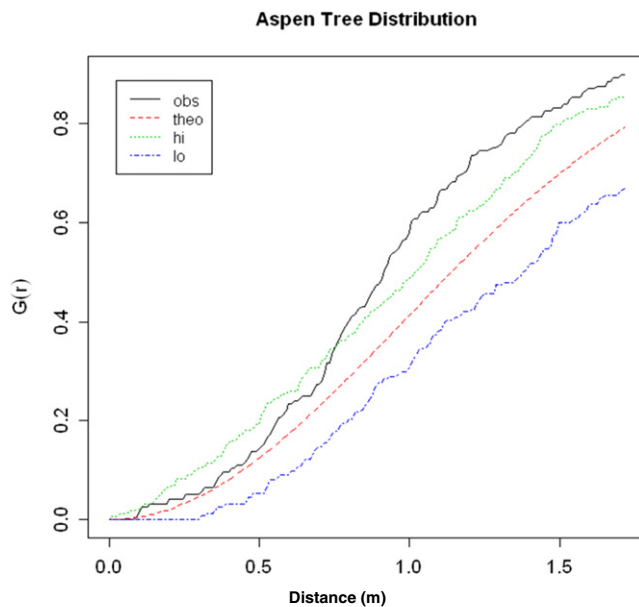


Fig. 2. Spatial distribution of aspen trees in a 14 year old oil sands reclaimed stand as estimated from the nearest-neighbor function $G(r)$. Observed $G(r)$ value above the simulation envelope indicates an aggregated pattern and within the envelope indicates a regular pattern (Obs = observed value of $G(r)$; Theo = theoretical value of $G(r)$; hi = upper envelop of $G(r)$; lo = lower envelop of $G(r)$).

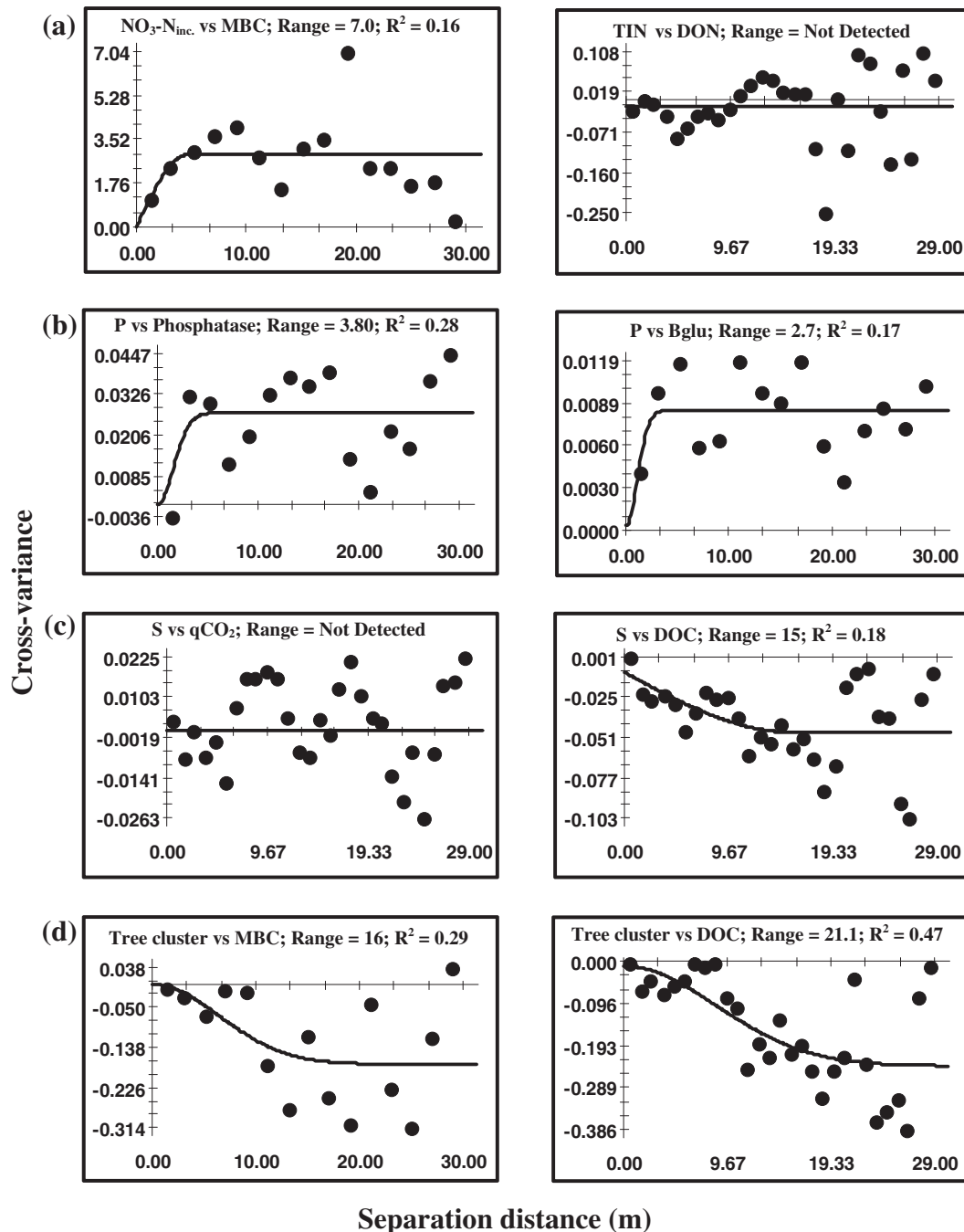


Fig. 3. Cross-variograms between available nutrients, soil properties, and stand characteristics of a 14 year old reclaimed aspen stand.

4.2. Spatial coupling between aboveground and belowground properties

Common spatial association between nutrient availability and soil microbial properties supports our second hypothesis. However, the spatial associations between soil microbial properties and tree cluster were either weak or occurred at a scale greater than 15 m, which is contrary to hypothesis 3, that these properties would be associated at the scale of tree clustering (~2–3 m; Fig. 2). Fine scale spatial association between nutrient availability and microbial properties indicates that belowground connections have been re-established in the reclaimed system. Spatial range of nutrients such as TIN, P and S had similar scale with most of the microbial properties (5 to 8 m), which suggests joint spatial processes at the given scale. The fine scale cyclic spatial association

between nutrients and microbial properties (<10 m) also supports the idea of re-established belowground connection. This type of periodicity might arise from evenly distributed interactions between two or more spatial processes (Bruckner et al., 1999), for example the coupling between MBC and labile C (HSC), as both of these properties showed a spatial range close to 10 m. Such regular distribution in belowground properties can result from the symmetric root and microbial competition for resources due to microsite variations, which ultimately determine the availability and spatial pattern of nutrients (Fajardo and McIntire, 2007; Raynaud and Leadley, 2005).

However, the expected cyclic pattern in aboveground attributes necessary to create regular microsite variations was not observed during our measurement. The weak and/or mostly large scale spatial

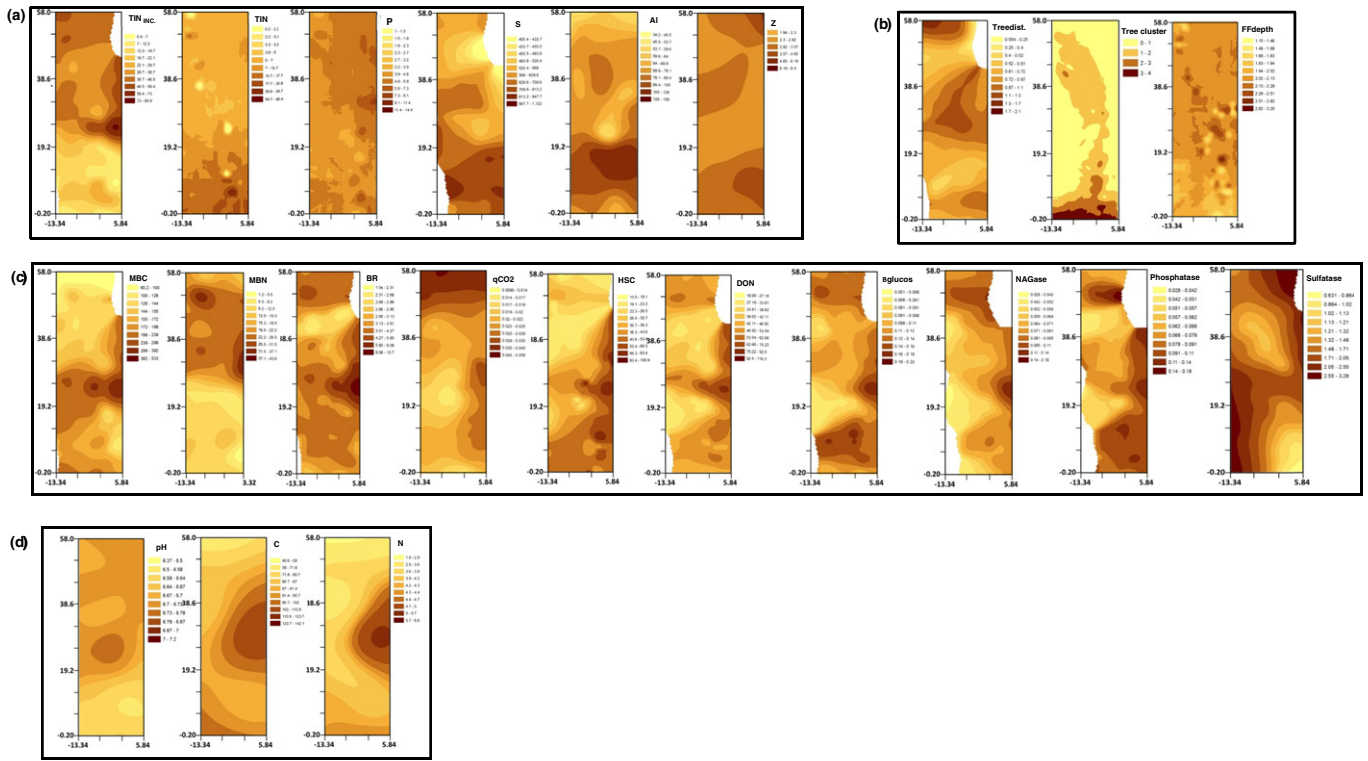


Fig. 4. Kriged contour map of TIN_{inc} and available nutrients measured by PRS probe (a), stand characteristics (b), soil microbial (c), and chemical (d) properties. X and Y axes indicates the coordinates of sample plot. Interpolation did not create complete map where contour lines were either too noisy or had island effect, and were removed from the maps as in the case of most of the microbial properties, S availability, and Treedist.

associations between stand characteristics and nutrient availability suggests that aboveground factors in the reclaimed aspen stand were not strong enough to directly contribute to heterogeneity in belowground

biological properties, at least in a spatial context. This is surprising given that a companion paper indicated that forest floor depth had a significant influence on microbial community structure as indicated by

Table 3

Standardized spatial regression coefficients for predictive variables of available macronutrients in a 14 year old aspen reclaimed stand (models shown in bold-italic text has significant spatial structure).

Properties	TIN			P			S			Ca			Mg		K	
	Full	WOSC ^a	WOMP ^b	Full	WOSC	WOMP	Full	WOSC	WOMP	Full	WOSC	WOMP	Full	WOSC	Full	WOMP
	Adj. R ²	Adj. R ²	Adj. R ²	Adj. R ²	Adj. R ²	Adj. R ²	Adj. R ²	Adj. R ²	Adj. R ²	Adj. R ²	Adj. R ²	Adj. R ²	Adj. R ²	Adj. R ²	Adj. R ²	Adj. R ²
	0.18	0.17	0.12	0.20	0.18	0.09	0.21	0.18	0.12	0.12	0.06	0.06	0.11	0.05	0.06	0.05
	AIC	AIC	AIC	AIC	AIC	AIC	AIC	AIC	AIC	AIC	AIC	AIC	AIC	AIC	AIC	AIC
	603	602	606	367	368	375	992	993	996	1070	1074	1073	807	810	830	829
Tree cluster	-	-	0.48*	-	-	-	-	-	0.11**	-	-	-	-	-	0.09	-
Treedist.	-	-	0.49	-	-	-	-	-	-	-	-	-	0.05	-	-	-
FFdepth	-	-	-	0.53	-	0.65	-	-	-	0.14*	-	0.14*	0.15*	-	-	-
FFmass	0.58	-	-	-	-	-	-	-	-	-	-	-	-	-	0.46*	0.48*
MBC	1.04*	0.87*	-	-	-	-	-	-	-	-	-	-	-	-	-	-
BR	-	-	-	-	-	-	-	-	-	0.05*	0.06*	-	-	-	-	-
HSC	-	-	-	-	-	-	0.33**	0.32**	-	-	-	-	-	-	0.16	-
PSC	-	-	-	-	-	-	-0.28*	-0.32**	-	-	-	-	-	-	-	-
DON	-	-	-	-	-	-	-0.49**	-0.44**	-	-	-	-	-	-	-	-
βglucosidase	-	-	-	0.96**	1.03**	-	-	-	-	-	-	-	-	-	-	-
NAGase	-	-	-	-	-	-	0.27*	0.19	-	-	-	-	-	-	-	-
Phosphatase	-	-	-	-0.36	-0.40	-	-	-	-	-	-0.04	-	-	-	-	-
Sulfatase	-1.86**	-1.87**	-	-	-	-	-0.09	-	-	-	-	-	-	-	-	-
C	-	-	-	0.58*	0.58*	-	-	-	-	-	-	-	-	-	-	-
N	-	-	-	-	-	0.65*	-	-	-	-	-	-	-	-	-	-
C:N	-	-	-	-	-	-	-	-	-	-	-	-	0.43*	0.34	-	-
pH	-18.6**	-18.2**	-15.9*	-	-	-	-	-	-	-	-	-	-1.73*	-1.62*	-	-
P	-	-	-	-	-	-	-0.11*	-	-0.09	-	-	-	-	-	-	-

"-" indicates variables were not selected in the spatial regression model.

^a WOSC = regression model without stand characteristics.

^b WOMP = regression model without microbial and chemical properties.

* p value < 0.05.

** p value < 0.005.

Table 4
Standardized spatial regression coefficients for predictive variables of available micronutrients in a 14 year old aspen reclaimed stand (models shown in bold-italic text has significant spatial structure).

Properties	Al			Mn			B			Cu		
	Full	WOSC^a	WOMP^b	Full	WOSC	WOMP	Full	WOSC	WOMP	Full	WOSC	WOMP
	Adj. R ²	Adj. R ²	Adj. R ²	Adj. R ²	Adj. R ²	Adj. R ²	Adj. R ²	Adj. R ²	Adj. R ²	Adj. R ²	Adj. R ²	Adj. R ²
	0.32	0.27	0.21	0.23	0.23	0.14	0.16	0.10	0.08	0.22	0.14	0.09
	AIC	AIC	AIC	AIC	AIC	AIC	AIC	AIC	AIC	AIC	AIC	AIC
	671	674	680	81	81	88	288	292	293	–175	–168	–165
Treedist.	-	-	-0.13*	-	-	-	-	-	0.18	-	-	-
FFdepth	0.33*	-	0.26	-0.86*	-	-0.92*	0.73*	-	0.81**	0.30**	-	0.30**
BR	-	-	-	0.72**	0.85**	-	0.38**	0.38**	-	0.08	0.08	-
HSC	0.44**	0.43**	-	-	0.53*	-	-	-	-	-	-	-
DON	-0.56**	-0.55**	-	-0.77**	-1.54**	-	-	-	-	-	-	-
βglucosidase	-	-	-	-0.45	-0.50	-	-	-	-	0.17*	0.18*	-
Phosphatase	-	-	-	-	-	-	-0.36*	-0.37*	-	-	-	-
C	-	-	-	-	-0.89*	-	-	-	-	-	-	-
N	-0.29*	-0.29	-	-	-	-	-	-	-	-	-	-
C:N	-	-	-	-	-	2.16*	-	-	-	0.55*	0.55*	-
pH	-5.1**	-5.2**	-4.8**	-10.9**	-9.4*	-9.4*	-	-	-	-	-	-1.29

“-” indicates variables were not selected in the spatial regression model.

^a WOSC = regression model without stand characteristics.

^b WOMP = regression model without microbial and chemical properties.

* p value < 0.05.

** p value < 0.005.

phospholipid analysis (Sorenson et al., 2011b). It is possible that the organic matter quality of the evenly distributed peat substrate is overriding the aboveground controls. A comparative study done in the AOSR found that sites reclaimed with peat have significantly different microbial biota and nutrient profiles than that of undisturbed natural analogues (Quideau et al., 2013). The organic matter structure of peat has also been shown to be different from that found in the forest floor of upland boreal ecosystems (Turcotte et al., 2009). Finally, a greenhouse study conducted by Pinno et al. (2012) showed that the performance of aspen seedlings was better on a salvaged forest floor substrate than a peat substrate similar to the one used here.

4.3. Mechanistic relationship between nutrient availability, soil and stand properties

Spatial regression models were used to account for the above and belowground control of nutrient availability in the reclaimed system. Strong soil microbial control on N–P–S availability in the regression models confirms that these nutrients are microbially mediated in the reclaimed environment (Table 3). This finding also supports our second hypothesis that microbial control on nutrient availability will be strong. Extracellular enzyme activity and labile C were the two important microbial predictors of nutrient availability besides soil pH. A few studies have reported similar observations of microbial control on nutrient availability in reclaimed ecosystems (Banning et al., 2008; Mummey et al., 2002); however they did not quantify the spatial relationships among these processes. We treated the spatial pattern of residuals separately in our regression models, and the resulting relationships between nutrient availability and predictor variables can be treated as mechanistic (Keitt et al., 2002).

We found a negative relationship between P availability and phosphatase activity in the spatial regression model. Negative feedback between P and phosphatase activity in previous studies was mostly attributed to the biotic demand for P (Juma and Tabatabai, 1978; Olander and Vitousek, 2000; Sinsabaugh et al., 1993; Tadano et al., 1993); however the actual mechanism of this relationship is not well understood. Goldstein et al. (1988) theorized that phosphatases are inducible enzymes dependent on end product inhibition. Plant roots and microbes do not usually exude phosphatase unless P availability does not meet their growth requirements. The negative relationship found

in this study and the fact that the peat substrate used for oil sand reclamation is low in available P (Pinno et al., 2012), represent evidence that biologically mediated belowground processes are trying to alleviate low P availability to satisfy growth requirements. Beta-glucosidase activity and TOC were the other two significant predictors of P availability as indicated by the regression model. Release of P from organic forms can either be a byproduct of C mineralization, where processes are mainly mediated by microbial energy (C) requirements, or can be a result of biochemical mineralization, where organisms are actively seeking P for biochemical reactions (McGill and Cole, 1981; Turner and Haygarth, 2005). Depolymerization of complex organic P polymers might require additional enzymes before the organic intermediates are available to phosphatases (McGill and Cole, 1981; Spears et al., 2001). Therefore, we can interpret the observed positive relationship between β-glucosidase activity and P availability to be the consequence of biochemical mineralization to cleave off the ester bound P (C–O–P) since the reclaimed system is P limited. A negative relationship was also detected between S and sulfatase activity in the spatial regression models. Similar to phosphatase, sulfatase is also subject to end product (SO_4^{2-}) inhibition. Several other studies reported a similar negative relationship between inorganic S and sulfatase activity (Maynard et al., 1985; Prietzel, 2001; Saviozzi et al., 2006). Maynard et al. (1985) found that sulfatase activity decreases when inorganic S concentration is high in soil solution. Wet deposition of S in the AOSR has been reported to be between 11 and 39 kg ha⁻¹ year⁻¹ (Proemse et al., 2012), which is much higher than the natural background of surrounding forested areas. Sulfate concentration in wet deposition often occurs in excess of vegetation requirements (Lindberg et al., 1986), likely moves through mineral soil, and this might have generated the inhibition of sulfatase activity in the study area. A positive relationship was observed between NAGase and S availability. This could happen when microbes use N rich compound as energy substrate to mineralize organically bound S (C–S or C–N–S) to meet C demand (McGill and Cole, 1981). This might be a possible mechanism of C mineralization in the reclaimed site, since S is already abundant in the system.

Although aboveground attributes in our study seem to have weak contribution to the availability of most of the essential macronutrients, forest floor depth showed a significant positive relationship with the availability of most base cations and micronutrients. The role of aspen as a cation pump has been recognized previously by different studies

(Hobbie et al., 2006; Paré and Bergeron, 1996). Returning cations to mineral soils through leaf litter is an important mechanism by which aspen might re-establish aboveground control on nutrient cycling and a return to natural function in the reclaimed environment. Aspen was previously found to initiate rapid forest floor development and establish links between aboveground and belowground processes in the AOSR (Sorenson et al., 2011a). However, development of spatial structure in micronutrients and base cations might take longer than the microbially mediated nutrients, as indicated by this study.

5. Conclusions

We used spatial analysis to tease apart scale dependent mechanistic relationships between nutrient availability, microbial properties and stand attributes 14 years after reclamation. This study revealed some interesting spatial control mechanisms of nutrient availability in the novel reclaimed ecosystem which might be useful for the management of such areas. Spatially structured variability in nutrient availability, microbial properties, and stand attributes indicated that the reclaimed ecosystem developed heterogeneous structure, although the spatial coupling between some of these properties was not well established. Stand attributes showed large scale spatial association and weak control on essential macronutrients (N, P and S) which suggest that availability of these nutrients in the reclaimed environment is more strongly regulated by belowground processes. In this young reclaimed system, N, P, and S availability was spatially predicted at less than 10 m scale from microbial properties such as extracellular enzyme activity and labile C. Availability of micronutrients, although not spatially structured, seemed to be controlled by forest floor thickness, soil microbial activity, and pH. Examining spatial patterns in ecosystem function will provide some of the data necessary to assess boreal forest reclamation, where interactions between many different woody perennial species and a living forest floor dominates. Future studies in the reclaimed ecosystems can also take advantage of the current spatial information to design ecologically meaningful sampling designs.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <http://dx.doi.org/10.1016/j.geoderma.2015.04.004>.

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